

EXP NO : 2

25/07/2026

DEVELOPING AND DEPLOYING A SIMPLE CHAINCODE

Aim:

To develop, package, install, approve, commit, and execute a simple asset transfer chaincode in a Hyperledger Fabric test network using the JavaScript programming language.

Requirements

- Kali Linux / Ubuntu
- Docker and Docker Compose
- Node.js (Version 18 or later) and npm
- Git
- Hyperledger Fabric Samples, Binaries, and Docker Images

Procedure & Execution

1. Navigate to the Test

Network Command(s):

```
(student@linux)-[~]  
$ cd ~/fabric-dev/fabric-samples/test-network
```

2. Start the Fabric Network

Command(s):

```
(student@linux)-[~]  
$ ./network.sh up createChannel -ca  
-----  
Creating channel 'mychannel'  
Starting peer0.org1  
Starting peer0.org2  
Starting orderer.example.com  
Network Started Successfully
```

3. Deploy the JavaScript

Chaincode Command(s):

```
(student@linux)-[~]  
$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript  
-----  
Packaging chaincode  
Installing chaincode  
Approving chaincode  
Committing chaincode  
Chaincode committed successfully
```

4. Verify Installed

Chaincode Command(s):

```
(student@linux)-[~]  
$ peer lifecycle chaincode queryinstalled
```

```
Package ID: basic_1.0  
Version: 1.0
```

5. Initialize the Ledger

Command(s):

```
(student@linux)-[~]  
$ export ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/  
orderer.example.com/msp/tlscacerts/tlsca.example.comcert.pem  
  
(student@linux)-[~]  
$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --  
cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'  
  
Chaincode invoke successful  
Transaction committed
```

6. Query All Assets

Command(s):

```
(student@linux)-[~]  
$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'  
  
[  
  {  
    "ID": "asset1",  
    "Color": "blue",  
    "Size": 5,  
    "Owner": "Tomoko",  
    "AppraisedValue": 300  
  },  
  {  
    "ID": "asset2",  
    "Color": "red",  
    "Size": 5,  
    "Owner": "Brad",  
    "AppraisedValue": 400  
  }  
]
```

7. Create a New Asset

Command(s):

```
(student@linux) - [ ]
```

```
└─$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com
```

NAME: **RENUGA DEVI K**

ROLL NO: **231901041**

NAME: **RENUGA DEVI K**

ROLL NO: **231901041**

```
["asset10","Black","15","User","800"]}]'
```

```
Asset asset10 created successfully
```

8. Query the Newly

Created Asset Command(s):

```
(student@linux)-[~]
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
-----
{
  "ID": "asset10",
  "Color": "Black",
  "Size": 15,
  "Owner": "User",
  "AppraisedValue": 800
}
```

9. Transfer Asset

Ownership Command(s):

```
(student@linux) - [ ]
$
```

```
Ownership transferred successfully
```

10. Verify Updated Asset

Command(s):

```
(student@linux)-[~]
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
-----
{
  "ID": "asset10",
  "Owner": "David"
}
```

11. Stop the Network

Command(s):

```
(student@linux)-[~]
$ ./network.sh down
-----
Stopping peer0.org1.example.com ... done
Stopping peer0.org2.example.com ... done
Stopping orderer.example.com ... done
Removing containers
Removing Docker Network
Network stopped successfully
```

Result:

The simple JavaScript chaincode was successfully developed, deployed, installed, approved, committed, and executed on the Hyperledger Fabric test network. Asset records were created, queried, and transferred successfully, demonstrating the basic lifecycle of smart contracts in Hyperledger Fabric.